Keeper Catch — "The 2D Substrate" Is Being Pinned to TWO Topologically Incompatible Objects (flat rank-2 T^2 vs curved color-S^2). Disambiguate by Role Before Grace Computes. Count 8.

Keeper | 2026-07-05 | Consistency catch across Grace + Lyra same-session resolutions. Not a gate — a disambiguation. Count 8.

The conflation

Two teammates pinned "the 2D substrate / base / floor" to two different objects in the same session; neither flagged the mismatch: - Grace: "the flat 2D floor — the rank-2 core T^2 is the leading candidate. Flat: no curvature." (Casey approved her pinning the floor to the rank-2 T^2.) - Lyra: "the 2D base is the color-sphere SO(3)/SO(2) = S^2 … the rank-2 core is a torus, not a sphere."

T^2 (genus 1, flat) and S^2 (genus 0, curved) are topologically distinct — they cannot be smoothed into one another. This is not a naming quibble; it changes what geometry gets computed.

The discriminant is Casey's own requirement

Casey: the commitment graph is a flat 2D surface that cannot collapse further. Flatness selects the torus and rules out the sphere. But the "sphere tiled with circles / Hopf base / clock-fiber / budding" picture needs the sphere. So each of Casey's two cosmology pictures wants a *different* 2D object.

Resolution: two objects, two roles (no contradiction as objects — only in the name)

Both live in the substrate's 5 = N_c(3 color) + rank(2) split, in orthogonal subspaces, so both exist simultaneously. The fix is to stop calling both "the 2D substrate":

object	subspace	curvature	role
rank-2 T^2	the 2 rank directions (strongly-orthogonal core)	flat	commitment floor — SWPP graph; "can't collapse"; Casey's flat surface
color S^2	the 3 color directions, SO(3)/SO(2)	curved	Hopf base — "sphere tiled with circles"; SO(2)-clock fiber; budding/spin

Why this is time-sensitive (the operational bite)

Grace is about to pin "the 2D floor to the rank-2 T^2" and compute its two circles. Lyra resolved the base to the color S^2. These are different objects: - Grace's T^2 computation serves the commitment floor, NOT the base-sphere. - The "compute-once, serve-three-hats" convergence (Grace: Shilov geometry → rung-2 + base + lepton kernel) assumes the base and the rung-2 Shilov live on one object. If the base is the color S^2 and rung-2 lives on the Shilov of the bulk, that may be two computations, not one. The elegant convergence could be a naming artifact.

Action: name the two objects separately (floor-T^2 vs base-S^2), then re-ask which one each target actually needs — rung-2 (Shilov of the bulk), the base (color S^2), the commitment floor (rank-2 T^2), Lyra's lepton kernel (which stratum?). Leave them fused and each gets computed on possibly-mismatched geometry.

Tier

Count 8. Both objects are solid, target-innocent group facts (T^2 = rank-2 core boundary; $S^2 = SO(3)/SO(2)$ on color 3-space). What is a candidate, not banked: which object is the commitment floor vs the Hopf base, and Lyra's load-bearing color→space descent (SO(3)_color → SO(3)_spatial) that would let the eternal color-S^2 become the emergent spin-S^2. The catch here is only that the two must be kept distinct; assigning roles is the team's forward call.

— Keeper, 2026-07-05. "The 2D substrate" = two topologically distinct objects: flat rank-2 T^2 (commitment floor, Casey's flat surface) and curved color-S^2 (Hopf base, sphere-tiled-with-circles). Genus 1 ≠ genus 0; flatness is the discriminant. Both exist in orthogonal subspaces of 5 = N_c + rank; disambiguate by role before computing, or the "compute-once" convergence may be a naming artifact. Count 8.